

A 10-MINUTE BRIEFING

From *Black-Box* to *Open*, Auditable AI.

Six ideas. One conclusion. Why the next phase of AI belongs
to open weights, small models, and auditable operations.

RISK → RISE → SMALL → MITIGATION → FUTURE

WHAT WE'LL COVER

01 02 03 04 05 06

~ 90 SECONDS PER SLIDE • QUESTIONS AT THE END

LET'S BEGIN —

Inside the transformer, the token journey goes *dark*.

WHAT ENTERS AS LANGUAGE EXITS
AS A GUESS — WITH NO RECEIPT

THE JOURNEY OF A TOKEN



No explainability

Attention ≠ business reasoning. The path from token to answer is opaque.



Hallucination

Confident, fluent — and often wrong. Closed APIs offer no source proof.



Data leakage

Prompts, docs, PII flow to a vendor with no audit of training reuse.



Silent model drift

Provider swaps weights overnight. Yesterday's controls no longer apply.



Lock-in & cost shock

Pricing, latency, and access change without warning. No fallback.

"Open" is a spectrum.
The world is racing *right* across it.

TIER	WEIGHTS	ARCH	TRAIN DATA	FINE-TUNE	SELF-HOST
Closed API e.g. GPT, Claude	×	×	×	×	×
Open Weight e.g. Llama, Mistral	✓	✓	×	✓	✓
Open Source e.g. OLMo, Pythia	✓	✓	✓	✓	✓

OPEN-WEIGHT ≠ OPEN-SOURCE. AUDIT TEAMS MUST READ THE LICENSE, NOT THE HEADLINE.

WHY IT'S RISING — FAST

- 70%+ of new enterprise pilots now test open-weight models alongside frontier APIs
- 10× cheaper inference on a fine-tuned 7B vs. a closed frontier API at scale
- 0 data leaves your VPC when models run on-prem or in a private cloud

Plus: *sovereignty, customization, no vendor lock-in*, and a thriving ecosystem of fine-tunes.

Open weights + small models + *agentic loop* = the new enterprise stack.

OPEN-SOURCE MODELS

Llama 4 META	17B-400B (MoE) Multimodal · long-context · reasoning
Mixtral 8x22B MISTRAL	141B / 39B active MoE reasoning · efficient inference
Qwen3 ALIBABA	0.6B-235B Multilingual · coding · agents
DeepSeek-V3 DEEPMATH	671B / 37B active Frontier reasoning · open MoE
Gemma 3 GOOGLE	1B-27B Lightweight · multimodal
Nemotron 3 Super NVIDIA	120B / 12B active Agentic · enterprise RAG
Command R+ COHERE	104B Tool-use · enterprise RAG
OLMo 2 A12	1B-32B Fully open (data + weights)

THE FOUNDATION

SMALL LANGUAGE MODELS

Phi-4 Complex reasoning at small scale	14B
Phi-3.5 mini Edge reasoning · on-device	3.8B
Mistral Small 3.1 Function calling · 128K context · vision	24B
Qwen3 Routing · extraction · multilingual	0.6 / 1.7 / 4 / 8B
Llama 3.2 Mobile · summarization · classification	1B / 3B
Gemma 3 On-device · multilingual	1B / 4B
SmolLM2 Embedded · ultra-low latency	135M - 1.7B
Mistral 7B General-purpose workhorse	7B

THE WORKHORSES

NEMO AGENT TOOLKIT

 **Perceive** *Sense the world*

- › Multi-modal input (text · vision · audio)
- › RAG over enterprise data
- › Tool & MCP server discovery

 **Think** *Plan & reason*

- › Cross-framework orchestration
- › Task decomposition & routing
- › Cost-aware model selection

 **Act** *Execute safely*

- › Tool calls via MCP
- › Guardrails · least-privilege scopes
- › Prompt-injection & jailbreak shields

 **Learn** *Improve continuously*

- › Evaluation & golden datasets
- › Cost · latency · accuracy tracking
- › Security testing & tool-poison checks

THE LOOP

WHY IT MATTERS *Agentic + SLMs* = 10× cheaper, on-prem, auditable digital workers — that *perceive, think, act, and learn* without leaking your data.

Open-source doesn't just *cut cost*. It closes the *trust gap*.

EACH BLACK-BOX RISK → AN OPEN-SOURCE ANSWER

No explainability	→	Inspect weights, attention, and reasoning traces — locally.
Hallucination	→	Ground every answer in your own RAG over your own data.
Data leakage	→	Run on-prem or in your VPC. Nothing leaves.
Silent model drift	→	Pin the exact weight hash. Upgrade only on your schedule.
Lock-in & cost shock	→	Swap providers and hardware without a rewrite.

WHAT COMES NEXT

- 01

Hybrid by default

Open-weight SLMs handle 80% of calls. Frontier APIs for the long tail.
- 02

Agentic + auditable

Every tool call logged, every model version pinned, every action replayable.
- 03

On-device intelligence

Phones, laptops, factories. Inference moves to where the data lives.
- 04

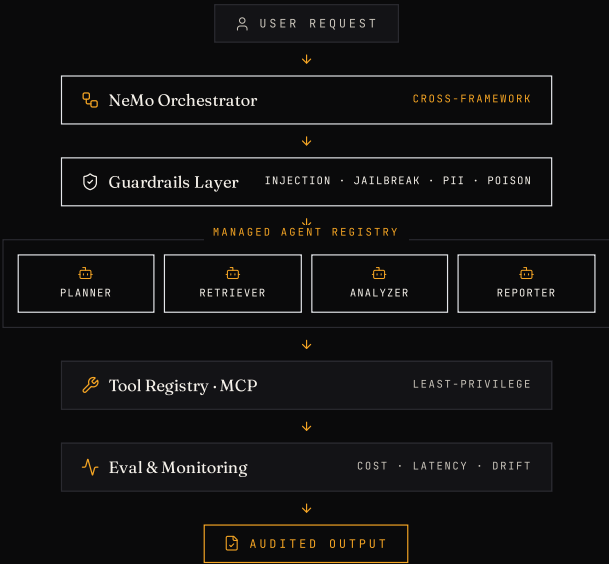
Open beats closed

Linux won servers. Open weights will win the AI layer — for the same reasons.

The winners will run a *portfolio* — not a single vendor.

How NVIDIA NeMo turns agents into *safe, governed* digital workers.

THE ARCHITECTURE



WHY IT'S EFFECTIVE

- Cross-framework**
LangChain · LlamaIndex · CrewAI · custom — onetoolkit
- Guardrails built-in**
Prompt-injection, jailbreak & tool-poison shields
- Managed agent registry**
Every agent versioned, scoped, and observable
- Eval at scale**
Golden datasets · accuracy · latency · cost
- FinOps native**
Per-agent, per-tool, per-task cost tracking
- Audit-ready**
Every prompt, tool call & output logged

⚡ THE RESULT

Agents that *perceive, think, act, and learn* — without going rogue.

From *black-box guesswork* to *open, auditable, agentic intelligence*.

THANK YOU · QUESTIONS WELCOME